

The Classical Baseline

Hartree-Fock, Full CI, and why the oracle comes before the thing it checks

Phase 1 writes no quantum code. It builds the exact classical reference that every subsequent phase is verified against, and develops the theory — mean-field approximation, Slater determinants, correlation energy, basis sets — that the quantum half of the project assumes you already have.

Estimated time: **1-2 hours**. The code is short; the theory is the substance.

In this document

Why writing VQE first is the most common way this project fails

The electronic Hamiltonian, and why nuclear repulsion is the term that creates the bond

Hartree-Fock as one Slater determinant; FCI as exact-within-a-basis

Correlation energy — the quantity VQE exists to capture

What STO-3G literally is, and where the four spin orbitals come from

Full build guide for `classical.py`, plus a decomposition check that catches inherited-Hamiltonian bugs

1. Where this phase sits

You are building the **oracle** — the trusted answer that every later phase gets checked against. No quantum code is written here either. That ordering is the single most important structural decision in the roadmap, and this section explains why.

The most common way this project fails

Write the VQE first. Get a number. Have no way to tell whether it is right.

A VQE that has converged to a local minimum, or that is using a Hamiltonian built at the wrong geometry, or that silently omitted the nuclear repulsion term, produces a plausible-looking curve that is simply wrong. It will have the right shape. It will have a minimum in roughly the right place. And it will be incorrect by an amount you cannot detect without an independent reference.

PySCF's FCI result for a four-spin-orbital system is **exact** and takes milliseconds. Build it first.

2. Theory

2.1 The electronic Hamiltonian

With the nuclei frozen at some geometry (Phase 0, Born-Oppenheimer), the operator whose lowest eigenvalue we want is:

$$\hat{H} = -\sum_i \frac{1}{2} \nabla_i^2 - \sum_{i,A} \frac{Z_A}{r_{iA}} + \sum_{i < j} \frac{1}{r_{ij}} + \sum_{A < B} \frac{Z_A Z_B}{R_{AB}}$$

term 1 *Electron kinetic energy.* Sum over electrons.

term 2 *Electron-nucleus attraction.* What binds electrons to the molecule. Z_A is the nuclear charge, 1 for hydrogen.

term 3 *Electron-electron repulsion.* The hard one — it couples every electron to every other and is why the problem does not factorise.

term 4 *Nuclear repulsion.* Depends only on the frozen nuclei, so at fixed geometry it is a plain constant. For H_2 it is simply $1/R$ in atomic units.

The first three terms involve electrons and are genuinely hard. The fourth is a constant — and because it is constant, it is frequently dropped, and because it is frequently dropped, it is frequently forgotten.

Nuclear repulsion is constant at fixed R — but you are scanning R

$1/R$ changes from 3.53 Ha at 0.3 Å to 0.42 Ha at 2.5 Å. Omit it and your curve has no minimum at all: the electronic energy alone falls monotonically as the atoms approach. **The nuclear repulsion term is what creates the bond.**

If you ever inherit a hardcoded Hamiltonian from a tutorial, the first thing to establish is whether this term is inside it. A 2-qubit H_2 Hamiltonian whose ground state is -1.857 Ha is electronic-only; one whose ground state is -1.137 Ha includes it.

2.2 The many-electron problem, and why it is hard

The difficulty is the electron–electron repulsion term. Because each electron's motion depends on where every other electron is, the wavefunction does not factorise into independent one-electron pieces. The exact solution requires considering every possible arrangement of electrons among orbitals simultaneously, and the number of such arrangements grows combinatorially.

The two classical methods you build in this phase sit at opposite ends of that difficulty.

2.3 Hartree-Fock: the mean-field approximation

Slater determinant	The simplest valid many-electron wavefunction: an antisymmetrised product of occupied spin orbitals. Antisymmetrised means it flips sign when you swap any two electrons, which is what fermions require. The Pauli exclusion principle falls out automatically.
Hartree-Fock (HF)	Approximate the true wavefunction by exactly one Slater determinant, and vary the orbitals to minimise the energy. Each electron then moves in the <i>average</i> field of all the others rather than dodging them individually.
Self-consistent field (SCF)	The iterative procedure that solves HF. The orbitals determine the average field; the average field determines the orbitals. Iterate until they stop changing.
Restricted HF (RHF)	The variant where paired electrons share the same spatial orbital. Correct for closed-shell molecules near equilibrium — and, as you will see in Phase 5, catastrophically wrong at long bond lengths.

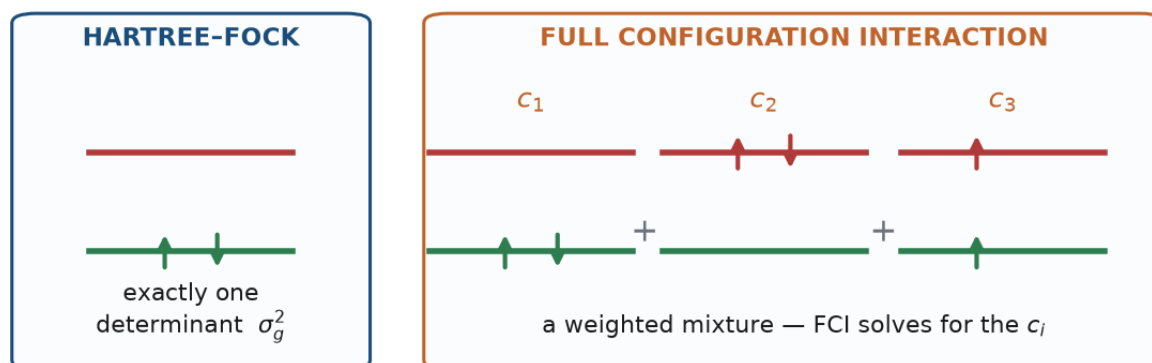
HF typically recovers around 99% of the total energy. That sounds excellent and is in fact useless on its own: chemistry lives in the remaining 1%, because bond energies and reaction barriers are differences between large numbers.

2.4 Full Configuration Interaction: exact, within a basis

FCI takes the opposite approach. Instead of one determinant, it uses a weighted mixture of **every** determinant that can be built from the available orbitals, and solves for the weights:

$$|\Psi_{\text{FCI}}\rangle = c_0|\Phi_{\text{HF}}\rangle + \sum_{ia} c_i^a |\Phi_i^a\rangle + \sum_{ijab} c_{ij}^{ab} |\Phi_{ij}^{ab}\rangle + \dots$$

One determinant versus a superposition of determinants



at equilibrium $c_1 \approx 1$ and HF is nearly right; as the bond stretches $c_2 \rightarrow c_1$ and HF collapses

Figure 1.1 — For H_2 in a minimal basis there are only a handful of determinants, which is exactly why FCI is tractable here and impossible for anything large.

"Exact" has a precise and limited meaning

FCI is the exact answer **within the chosen basis set**. It is not the exact answer to reality. FCI in a minimal STO-3G basis is the exact solution of a deliberately simplified question.

This is why you compare VQE against *your own FCI in your own basis*, and never against an experimental number from the literature.

2.5 Correlation energy

$$E_{\text{corr}} \equiv E_{\text{FCI}} - E_{\text{HF}}$$

This is the part mean-field theory misses — the energy lowering that comes from electrons genuinely avoiding one another rather than each responding to an average. At equilibrium for H_2 it is about 0.0203 Ha, or 0.55 eV. **It is the entire reason VQE exists.** A quantum algorithm that could not capture correlation energy would have no purpose in chemistry.

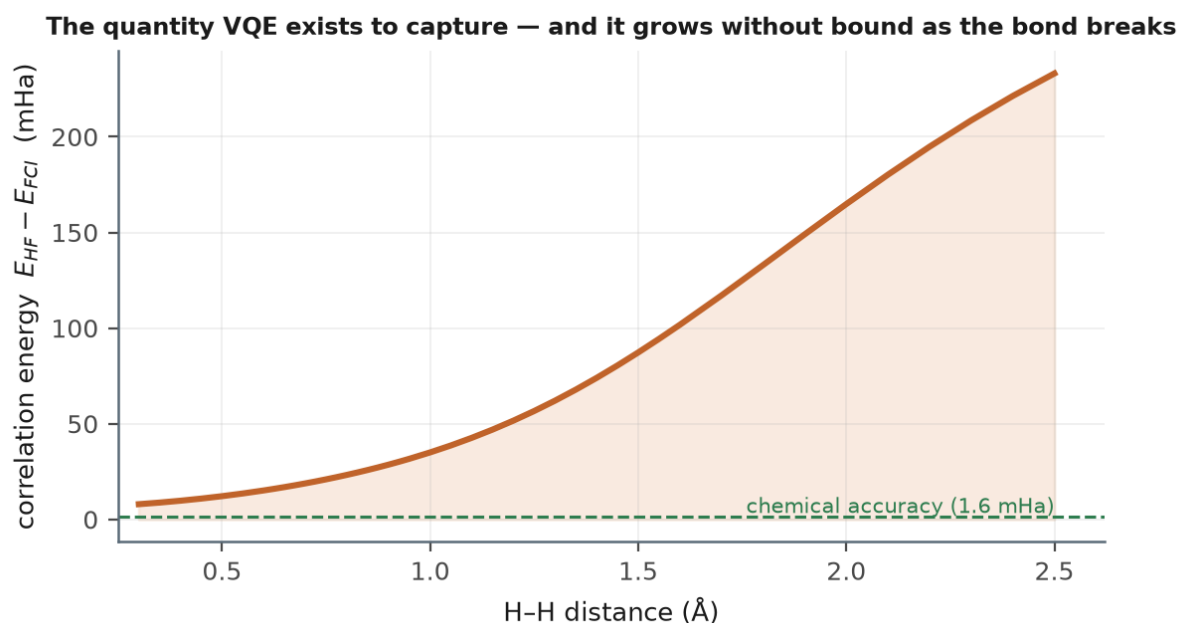


Figure 1.2 — Correlation energy against bond length, from the reference run. Note that it does not stay small.

The growth on the right of that plot is the important feature. Near equilibrium correlation is a small correction; as the bond breaks it becomes the dominant term. That regime is called **static correlation**, and Phase 5 shows you exactly what it looks like.

2.6 Basis sets, and what STO-3G literally is

Orbitals must be expressed as combinations of mathematical functions, and the chosen menu of functions is the **basis set**. The physically correct shape for a hydrogen 1s orbital is a Slater function, e^{-r} , but Slater functions make the two-electron integrals analytically intractable. Gaussians, $e^{-\alpha r^2}$, have easy integrals but the wrong shape.

STO-3G is the compromise: approximate each **Slater-Type Orbital** by **3** fitted **Gaussians**. It is the smallest usable basis, which is why it is called *minimal* — each atom gets the bare minimum of orbitals and nothing more.

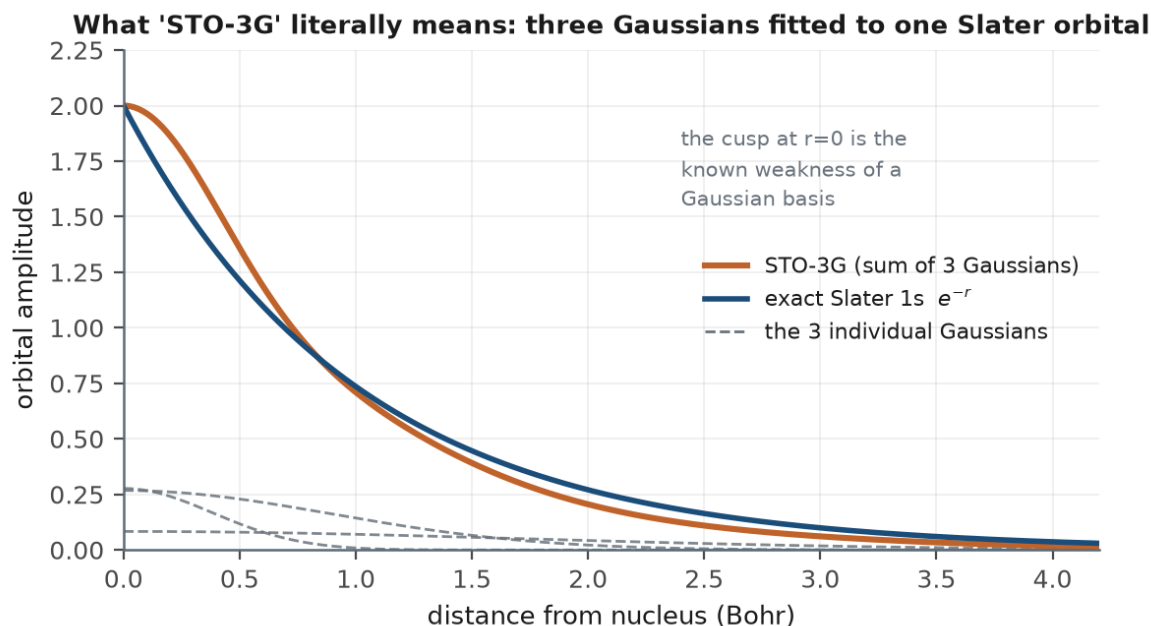


Figure 1.3 — STO-3G is a literal curve fit. The three dashed Gaussians sum to the orange curve, which approximates the blue Slater orbital.

For H_2 this gives one basis function per atom, so two spatial orbitals: the bonding σ_g and the antibonding σ_u^* . Each holds two electrons of opposite spin, giving **four spin orbitals** — the number that becomes four qubits in Phase 2.

H_2 molecular orbitals and the four spin orbitals

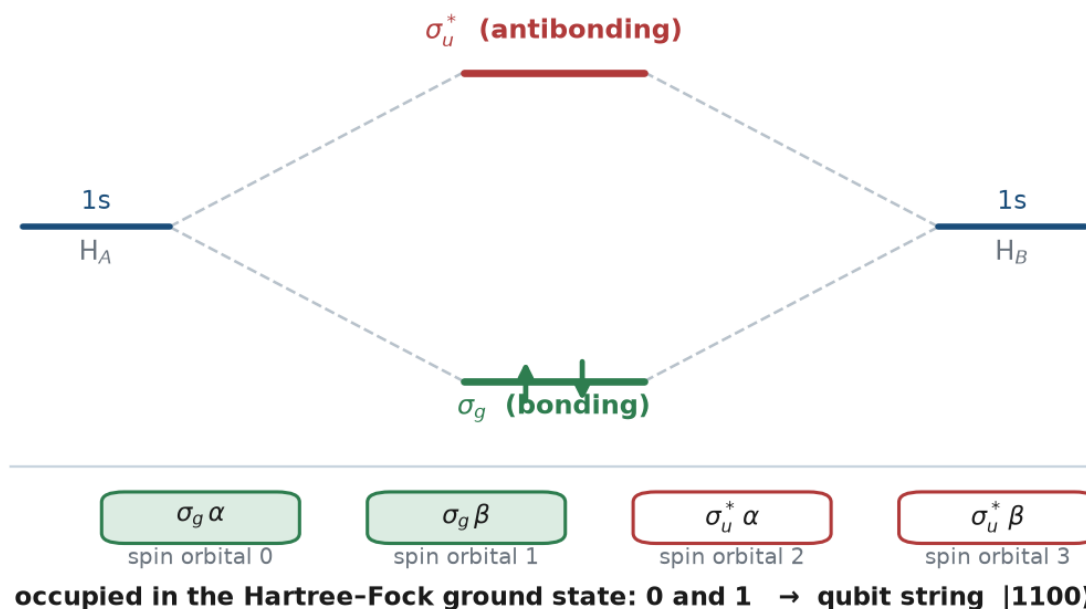


Figure 1.4 — The four spin orbitals, and the Hartree-Fock occupation that becomes the VQE reference state in Phase 3.

3. Build guide

Step 1.1 — A single geometry, both methods

Create `classical.py`. Start with the simplest possible thing that produces a checkable number.

Listing 1.1 — classical.py

```
"""classical.py - Hartree-Fock and FCI reference energies from PySCF."""

from pyscf import gto, scf, fci

def classical_energies(r_angstrom, basis="sto-3g"):
    """Return (E_HF, E_FCI, E_nuclear_repulsion) for H2 at bond length r.

    All energies are TOTAL energies in Hartree, i.e. they already include
    the nuclear repulsion term. This matters - see Phase 1, section 2.1.
    """
    mol = gto.M(
        atom=f"H 0 0 0; H 0 0 {r_angstrom}",
        basis=basis,
        unit="Angstrom",      # be explicit; the default is Angstrom but
        verbose=0,            # PennyLane's default is Bohr. Never assume.
    )

    mf = scf.RHF(mol)
    e_hf = mf.kernel()

    e_fci = fci.FCI(mf).kernel()[0]

    return e_hf, e_fci, mol.energy_nuc()

if __name__ == "__main__":
    r = 0.735
    e_hf, e_fci, e_nuc = classical_energies(r)

    print(f"bond length           {r:12.4f} Angstrom")
    print(f"nuclear repulsion       {e_nuc:12.6f} Ha")
    print(f"Hartree-Fock                {e_hf:12.6f} Ha")
    print(f"FCI                        {e_fci:12.6f} Ha")
    print(f"correlation energy         {e_hf - e_fci:12.6f} Ha "
          f"f"({(e_hf - e_fci) * 27.2114:.4f} eV)")
```

Step 1.2 — Run it and check against the reference

```
$ python classical.py
bond length           0.7350 Angstrom
nuclear repulsion     0.719969 Ha
Hartree-Fock          -1.116999 Ha
FCI                   -1.137306 Ha
correlation energy     0.020307 Ha (0.5526 eV)
```

Those four numbers are the anchors for the entire project. Write them down. Every later phase is checked against the FCI value, and if you ever see a VQE energy *below* -1.137306 Ha at this geometry, you have a bug — the variational principle forbids it (Phase 3).

Step 1.3 — Verify the pieces add up

A useful habit: confirm that the total energy really does decompose the way you think it does. Add this check.

Listing 1.2 — a decomposition sanity check

```
def decomposition_check(r_angstrom=0.735):
    """Confirm that total = electronic + nuclear repulsion."""
    mol = gto.M(atom=f"H 0 0; H 0 0 {r_angstrom}", basis="sto-3g",
                unit="Angstrom", verbose=0)
    mf = scf.RHF(mol)
    e_total = mf.kernel()
    e_nuc = mol.energy_nuc()
    e_electronic = e_total - e_nuc

    print(f"electronic only      {e_electronic:12.6f} Ha")
    print(f"nuclear repulsion    {e_nuc:12.6f} Ha")
    print(f"total                  {e_total:12.6f} Ha")

    # 1/R in atomic units, computed by hand as an independent check.
    # Tolerance is 1e-6, not 1e-12: PySCF and this file use CODATA
    # constants from different revisions, which differ in the 10th
    # significant figure.
    BOHR = 1.8897261254578281
    expected_nuc = 1.0 / (r_angstrom * BOHR)
    assert abs(e_nuc - expected_nuc) < 1e-6, "nuclear repulsion mismatch"
    print(f"hand-computed 1/R      {expected_nuc:12.6f} Ha   OK")
    print(f"    (differ by {abs(e_nuc - expected_nuc):.2e} Ha - "
          f"different CODATA revisions, not a bug)")

electronic only      -1.836968 Ha
nuclear repulsion    0.719969 Ha
total                -1.116999 Ha
hand-computed 1/R    0.719969 Ha   OK
(differ by 3.40e-10 Ha - different CODATA revisions, not a bug)
```

That residual is worth a word. It is **not** a numerical error in your calculation — it is the difference between the Bohr-to-Ångström conversion PySCF uses internally and the one you wrote. Set the tolerance to 10^{-12} and this check fails for a reason that has nothing to do with physics.

Why this check earns its place

It is the check that catches an inherited Hamiltonian missing its nuclear repulsion term. If you ever compare against a published 2-qubit H_2 operator, its ground state will be around -1.857 Ha; add $1/R = 0.720$ Ha and you land on -1.137 Ha. Knowing this decomposition means you can reconcile the two rather than wondering which is wrong.

Step 1.4 — Scan the geometry classically

Before any quantum code exists you can already produce two thirds of the final figure. This is worth doing now: it validates your grid, gives you the shape you are aiming for, and takes seconds.

Listing 1.3 — the classical scan

```
import numpy as np

def classical_scan(grid, basis="sto-3g"):
```



```
"""HF and FCI across a range of bond lengths."""
e_hf = np.zeros(len(grid))
e_fci = np.zeros(len(grid))

for i, r in enumerate(grid):
    e_hf[i], e_fci[i], _ = classical_energies(r, basis)
    print(f" r={r:5.3f} HF={e_hf[i]:9.6f} FCI={e_fci[i]:9.6f}")

return e_hf, e_fci

if __name__ == "__main__":
    grid = np.arange(0.3, 2.55, 0.05).round(4)
    e_hf, e_fci = classical_scan(grid)
    np.savez("data/classical_scan.npz", grid=grid, e_hf=e_hf, e_fci=e_fci)
```

Plot those two curves immediately. You are looking for: a clear minimum near 0.735 Å in both; the FCI curve flattening out on the right; and the HF curve **failing** to flatten. That divergence is the subject of Phase 5 and it is the most instructive feature you will produce.

4. Traps in this phase

Trap	Symptom	Fix
Unit left implicit	Minimum appears at a physically absurd distance, or there is no minimum at all.	Pass <code>unit="Angstrom"</code> explicitly to <code>gto.M</code> . PySCF defaults to Ångström but PennyLane defaults to Bohr — being explicit everywhere prevents the Phase 2 version of this bug.
Comparing to experiment	Well depth and bond length disagree with textbook values.	Expected. STO-3G gives poor absolute energies. Compare against your own FCI in the same basis.
SCF fails to converge	PySCF warns about convergence at large R.	RHF genuinely struggles as the bond breaks — this is physics, not a numerical fault. It is the same failure Phase 5 analyses. Raise <code>mf.max_cycle</code> if needed, but expect it.
Forgetting nuclear repulsion	The curve has no minimum; energy falls monotonically as $R \rightarrow 0$.	Use <code>mf.e_tot</code> (total) rather than the electronic energy. Listing 1.2 catches this.
Trusting <code>verbose=0</code> output	Silent convergence failure.	Check <code>mf.converged</code> is <code>True</code> before using the result.

5. What was done in this phase

- **Built the oracle first.** You now have an independent, exact reference for every geometry, produced by a completely different method from the one you are about to test. This is the only thing that makes Phases 3 through 6 verifiable rather than merely plausible.
- **Established the electronic Hamiltonian** and, crucially, the fact that the nuclear repulsion term is what creates the bond — and the term most often silently missing from inherited Hamiltonians.
- **Understood Hartree-Fock** as a single-Slater-determinant mean-field approximation, and **FCI** as the exact solution within a basis, along with the precise and limited sense in which "exact" applies.
- **Defined correlation energy** as the gap between them — 0.0203 Ha at equilibrium — and saw that it grows without bound as the bond breaks.
- **Understood what STO-3G literally is:** three Gaussians fitted to one Slater orbital, giving two spatial orbitals and therefore four spin orbitals for H_2 .
- **Produced two thirds of the final figure** before writing a single line of quantum code.

6. Checklist before moving on

1. `classical.py` prints HF = −1.116999 Ha and FCI = −1.137306 Ha at 0.735 Å.
2. The decomposition check passes: electronic + 1/R = total.
3. You have a classical scan saved to disk and plotted.
4. The HF curve visibly fails to flatten at large R while FCI does.

5. You can state why comparing STO-3G results to experiment is a category error.

Next

Phase 2 — The qubit Hamiltonian. This is the stage most tutorials skip by handing you hardcoded Pauli coefficients. You will build it properly: second quantisation, the fermion-to-qubit mapping, and the symmetry tapering that turns four qubits into two.